MULTIMEDIA UNIVERSITY ®

Faculty of Information Science and Technology (FIST)

FIST-FYP Project Proposal Form

Instructions:

This form is for students to submit their Final Year Project (FYP) proposal.
Please complete this proposal using the provided template and send it to your respective supervisor for review. Once reviewed, it should be submitted to the FYP Committee via the FIST-FYP website for approval. Fields marked with * are compulsory. Kindly read the notes at the bottom of this document before writing your proposal.

Project Title *	Intelligent Campus Assistant Chatbot using RAG Technology
Description *	This is a project that builds up an **AI-based Campus Assistant** to help students in navigate fragmented university information. This system uses **Retrieval-Augmented Generation (RAG)** to base answers on official information unlike the normal chatbots which allow AI hallucinations. The core technical input consists in the engineering of a **Hybrid Knowledge Base:** it reads unstructured text of official MMU Handbooks (PDFs) containing academic regulations and incorporates a self-built structured dataset of campus location navigability. The system applies Vector Search to the retrieval of specific context (e.g., of specific rules or areas of a room) and creates natural language responses through a Large Language Model (LLM).
Objectives *	• To develop a chatbot pipeline based on RAG to extract and synthesize information on both unstructured student handbooks and structured campus location data. • To engineer a custom vector-indexed knowledge base that consolidates academic regulations, facility details, and staff directory information. • To determine the accuracy of the system in retrieving as well as the relevancy of responses against the conventional method of search using keywords.
Project type *	Application-based
Specific area of research *	Natural Language Processing (NLP), Large Language Models (LLM), Information Retrieval & Data Engineering
Skill set required *	Python, LangChain Framework, Vector Databases (ChromaDB), Data Engineering (PDF Parsing/JSON structuring), LLM APIs
Expected deliverable *	An effective web-based chatbot prototype that may answer academic and navigational questions with a performance evaluation report.
Project schedule	1. Literature review on RAG architectures and Data Engineering (1 month) 2. Data Collection & Preprocessing (Scraping PDFs + Building Location Dataset) (1 month) 3. System Development (Vector Indexing & Chat UI) (3 months) 4. System Testing & Accuracy Evaluation (1 month) 5. Deployment & Final Report writing (1 month)

Innovative component/ project potential	Enhancement: replaces static FAQ lists with a dynamic **Semantic Search** engine that understands context (e.g., distinguishing between "Faculty of Business" and "FIST" rules).
This project suitable for programme * Please ensure the project is suitable for the programme you select here and ensure also it is link with the programme outcome.	[ ] Data Communications and Networking [ ] Security Technology [X] Artificial Intelligence [ ] Business Intelligence and Analytics [ ] Bionformatics

Note: All the proposals submitted will be review by FYP committee/ reviewers using *double-blinded* reviewing system. Thus, to avoid rejection or revision of proposal, kindly ensure you consider the following questions and preferably "YES" and provide with good justification.

1. Is the FYP proposal relevant to the selected programme?

   YES. The project is at the centre of **Artificial Intelligence and Data Science**. It is devoted to **Natural Language Processing (NLP)** and Information Retrieval and presents the use of **Retrieval-Augmented Generation (RAG)** to address a problem in a specific area. It also entails data engineering (organizing non-structured PDF and location data) which is the basis of developing AI applications.

2. Is the project sufficient for Final Year Student level?

   YES. This project goes beyond a simple CRUD (Create-Read-Update-Delete) application. It involves complex architectural components including **Vector Database indexing**, **Semantic Search implementation**, and **LLM integration**. The necessity to create a special hybrid data (unstructured text with structured location data) brings a lot of engineering value that could be used in an undergraduate thesis.

3. Are the project's title and description clear and sound?

   YES. The title clearly indicates the technology **RAG** is used and the field of application **Campus Assistant**. It is explicitly stated in the description that the present issue is information fragmentation, and that a particular technical remedy (AI-based semantic search) within certain limits (MMU campus setting) is being proposed.

4. Are the project's objectives significant and creditable?

   YES. The objectives are specific, measurable, and technically challenging. They cover the full development lifecycle: from **Data Engineering** (building the location dataset) to **System Implementation** (RAG pipeline) and **Performance Evaluation** (accuracy testing). The project solves a real-world problem for the university community, giving it high practical significance.

5. Will the student be able to complete the project within 6-7 months?

   YES. The scope is well-defined and manageable.
   - **Months 1-2:** Data collection (PDFs) and cleaning.
   - **Months 3-5:** Development using rapid prototyping tools (Python Streamlit, LangChain) which significantly reduce coding time compared to traditional web stacks.
   - **Months 6-7:** Testing and documentation. The use of existing APIs (Gemini/OpenAI) eliminates the need for time-consuming model training from scratch.

6. Are the major skill-sets required covered in the programme courses?

   YES. The project will be based on the Python programming, Database Management, and Software Engineering principles, which form part of the core elements of the curriculum. It takes some of the basics of algorithms and data structures and applies them to a current AI problem.

7. Is the student required to learn new concepts, knowledge or techniques to complete the project?

   YES. Although the groundwork is already covered, I will have to study and apply modern AI technologies that are not completely included in the standard coursework and that are: **Vector Embeddings, LangChain orchestration, and Prompt Engineering**. This shows the capability to do an independent research and use a high-end technology.